

PU-II BIOLOGY · CHAPTER 7: HUMAN HEALTH & DISEASE (BOARD QUESTION BANK)

★ 100% GUARANTEED BOARD QUESTIONS

PART A · VERY SHORT ANSWER QUESTIONS (1 MARK EACH)

Q1. Define Health according to WHO.

1 Mark

Health is defined as a state of complete **physical, mental, and social well-being**, and not merely the absence of disease or infirmity.

Q2. Name the confirmatory diagnostic test for Typhoid.

1

Mark Repeated

Widal test (serological test).

Q3. Name the toxic substance responsible for chills in malaria.

1

Mark Repeated

Hemozoin (released upon the rupture of infected red blood cells).

Q4. What is the infectious stage of *Plasmodium* to humans?

1 Mark

Sporozoite (injected via salivary secretions of female *Anopheles*).

Q5. Which type of antibody is present abundantly in colostrum?

1

Mark Repeated

Secretory IgA antibodies.

Q6. Which antibody mediates allergic reactions?

1 Mark

IgE antibodies.

Q7. What are Interferons? State their defensive role.

1

Mark Repeated

Interferons are **cytokine glycoproteins** secreted by virus-infected cells that protect neighboring non-infected cells from further viral invasion.

Q8. Expand MALT and state its share in lymphoid tissue.

1 Mark

Mucosa-Associated Lymphoid Tissue; constitutes approximately **50%** of the total lymphoid tissue in the human body.

Q9. Which immune response causes graft rejection?

1

Mark Repeated

Cell-Mediated Immune Response (CMIR) mediated by T-lymphocytes.

Q10. What is metastasis?

1 Mark Repeated

The sloughing of cancer cells from malignant tumors that migrate via blood/lymph to seed secondary tumors at distant organs.

Q11. Name the plant source of Cocaine and its mode of action.

1 Mark

Erythroxylum coca; interferes with the transport/reuptake of the neurotransmitter **Dopamine**.

Q12. What chemical compound is Heroin chemically?

1 Mark

Diacetylmorphine (obtained by the acetylation of morphine).

PART B · SHORT ANSWER QUESTIONS (2 MARKS EACH)

Q13. Differentiate between Benign and Malignant tumors.

2

Marks Very Imp

- **Benign:** Remain confined to origin; encapsulated; non-invasive; cause little damage.
- **Malignant:** Masses of proliferating neoplastic cells; rob nutrients; exhibit **metastasis** to distant body tissues.

Q14. What are Allergens? Name two chemicals released.

2 Marks

Allergens are environmental substances that elicit an exaggerated immune response (e.g., pollen, dust mites).
Chemicals released by mast cells: **Histamine** and **Serotonin**.

Q15. Mention the pathogen, site of infection, and vector of Amoebiasis.

2 Marks

- **Pathogen:** *Entamoeba histolytica* (protozoan).
- **Site:** Large intestine (mucosa/submucosa).
- **Mechanical Vector:** Housefly (*Musca domestica*).

Q16. What is contact inhibition? How do cancer cells differ?

2

Marks Very Imp

Contact inhibition is the property of normal cells where contact with neighboring cells inhibits uncontrolled proliferation.
Cancer cells lose contact inhibition and divide continuously into tumors.

PART C • ANALYTICAL & DISTINGUISHING QUESTIONS (3 MARKS EACH)

Q17. Differentiate between Innate Immunity and Acquired Immunity.

3 Marks Repeated

Feature	Innate Immunity	Acquired Immunity
Occurrence	Congenital; present by birth	Acquired during lifetime upon pathogen contact
Specificity	Non-specific defense mechanism	Highly pathogen and antigen specific
Memory	Lacks immunological memory	Characterized by lifelong immunological memory
Components	Physical, physiological, cellular & cytokine barriers	B-lymphocytes (antibodies) and T-lymphocytes

Q18. Differentiate between Active Immunity and Passive Immunity with examples.

3 Marks 100% Expected

Parameter	Active Immunity	Passive Immunity
Antibody Production	Host's own B-cells produce antibodies in response to antigen.	Pre-formed readymade antibodies directly administered to host.
Onset & Duration	Slow onset with latency phase; long-lasting protection with memory.	Immediate action; short-lived protection; no memory developed.
Natural Example	Recovery following natural measles or chickenpox infection.	Maternal IgG across placenta; Secretory IgA via colostrum.
Artificial Example	Vaccines (e.g., Polio, BCG, Hepatitis B, MMR).	Anti-tetanus serum (ATS); Anti-venom for snakebites.

Q19. Describe the four non-specific barriers of Innate Immunity.

3 Marks Repeated

- **1. Physical Barriers:** Skin (keratinized epidermis) blocks entry of microbes; mucus coating in respiratory, gastrointestinal, and urogenital tracts traps incoming microbes.
- **2. Physiological Barriers:** Hydrochloric acid (HCl) in gastric juice, lysozyme in tears and saliva, and sebum inhibit and destroy microbial growth.
- **3. Cellular Barriers:** Phagocytic leukocytes like PMNL-neutrophils, monocytes, and tissue macrophages engulf microbes; Natural Killer (NK) cells destroy abnormal cells.
- **4. Cytokine Barriers:** Virus-infected cells produce **interferons** (glycoproteins) that diffuse to neighboring healthy cells, inducing resistance to viral replication.

Q20. Primary vs. Secondary Lymphoid Organs. 3 Marks Very Imp

- **Primary Lymphoid Organs:**
 - Organs where immature lymphocytes differentiate and mature into immunocompetent cells.
 - *Examples:* **Bone marrow** (produces all blood cells; B-cells mature here) and **Thymus** (T-cells mature here).
- **Secondary Lymphoid Organs:**
 - Organs where mature lymphocytes encounter antigens, undergo clonal proliferation, and generate immune responses.
 - *Examples:* **Spleen, Lymph nodes, Tonsils, Peyer's patches, and MALT.**

Q21. Explain Withdrawal Syndrome & Dependence. 3 Marks

- **Dependence:** The physiological and psychological adaptation of the body to regular drug/alcohol intake, such that normal functioning requires the substance.
- **Withdrawal Syndrome:** Manifestation of distressing symptoms when regular drug intake is abruptly discontinued.
 - *Key Symptoms:* Severe anxiety, tremors/shakiness, profuse sweating (diaphoresis), nausea, vomiting, and palpitations.
 - Relieved upon resuming intake; requires clinical detoxification.

PART D • LONG ESSAY & DIAGRAMMATIC QUESTIONS (5 MARKS EACH)

Q22. Describe the Life Cycle of *Plasmodium* in Human and Mosquito Hosts.

5 Marks 100% Guaranteed 5-Marker

Plasmodium is a digenetic parasite requiring two hosts: Human (secondary host) and Female *Anopheles* mosquito (primary host).

- **1. Infection Phase:** Female *Anopheles* bites a healthy human, injecting infectious, spindle-shaped **sporozoites** via its saliva into the bloodstream.
- **2. Pre-erythrocytic (Hepatic) Schizogony:** Sporozoites reach the liver within 60 minutes, enter liver hepatocytes, reproduce asexually, and burst liver cells to release **merozoites** into the bloodstream.
- **3. Erythrocytic Schizogony:** Merozoites penetrate red blood cells (RBCs), pass through trophozoite stages, and reproduce asexually. The mature schizonts burst the RBCs, releasing merozoites and the toxic crystalline pigment **hemozoin** (responsible for the characteristic cyclic chills and spiking high fever).
- **4. Gametocytogenesis:** A fraction of intra-erythrocytic merozoites differentiate into sexual stages: male (σ) microgametocytes and female (φ) macrogametocytes.
- **5. Mosquito Phase (Sporogony):** When a female *Anopheles* bites the infected person, it ingests gametocytes along with blood meal. In the **mosquito midgut**, fertilization occurs forming a diploid motile **ookinete**. The ookinete penetrates the stomach wall, forming an encysted **oocyst**. The oocyst undergoes sporogony (nuclear fission), producing thousands of **sporozoites** that burst out and migrate to the **salivary glands**, completing the cycle.

Q23. With a neat labeled schematic, describe the structure of an Antibody Molecule.

5 Marks Extremely Common 5-Marker

Antibodies are immunoglobulin (Ig) glycoproteins synthesized by B-lymphocyte plasma cells.

- **Molecular Formula (H_2L_2):** Each antibody monomer consists of **4 polypeptide chains**:
 - **Two identical Heavy (H) chains:** Longer chains (~440 amino acids, ~50 kDa).
 - **Two identical Light (L) chains:** Shorter chains (~220 amino acids, ~25 kDa).
- **Disulfide Bridges:** Chains are held in a 'Y' configuration by covalent inter-chain and intra-chain **disulfide (S-S) bonds**.
- **Variable (V) Region (F_{ab} Fragment):** Located at the amino-terminal tips (V_H and V_L). Forms the **antigen-binding site (paratope)** that binds specifically to the epitope on the antigen in a lock-and-key manner. Each antibody has **2 antigen-binding sites** (bivalent).
- **Constant (C) Region (F_c Fragment):** Carboxy-terminal stem domains that mediate biological effector functions, macrophage opsonization, and complement fixation.

Q24. Explain the Pathogenesis, Replication Cascade, and Diagnosis of HIV / AIDS.

5 Marks Very Imp 5-Marker

HIV (Human Immunodeficiency Virus): Enveloped retrovirus with two single-stranded RNA (+ssRNA) filaments and Reverse Transcriptase.

- **Step 1 (Attachment & Entry):** Viral glycoprotein spikes (gp120) bind specifically to **CD4 receptor** markers on host macrophages and Helper T-lymphocytes (T_H).
- **Step 2 (Reverse Transcription):** The enzyme **Reverse Transcriptase** transcribes viral RNA into a complementary molecule of viral DNA inside host cytoplasm.
- **Step 3 (Integration & HIV Factory):** Viral DNA enters the nucleus and integrates into the host chromosome via the enzyme **Integrase**. Host cell transcribes viral RNA. The infected **macrophage survives and continuously produces new viral particles** ("HIV Factory").
- **Step 4 (T_H Cell Lysis & Immunodeficiency):** HIV simultaneously enters Helper T-cells ($CD4^+ T_H$), multiplies, and lyses them. Progressive depletion of T_H cells below critical levels (<200 cells/ μ L) cripples cell-mediated immunity.
- **Step 5 (Opportunistic Invasions):** The patient succumbs to severe opportunistic pathogens: *Mycobacterium*, *Toxoplasma*, CMV, and fungal infections.
- **Diagnosis:** **ELISA** (Enzyme-Linked Immunosorbent Assay) for screening; **Western Blotting** for confirmation.

Q25. Write an essay on Abused Drugs: Opioids, Cannabinoids, and Coca Alkaloids.

5 Marks Standard 5-Marker

- **1. Opioids (Heroin / Smack):** Extracted from latex of opium poppy (*Papaver somniferum*). Heroin is chemically diacetylmorphine (white, bitter, odorless crystalline solid). Mode of action: Binds specific opioid receptors in the CNS and gastrointestinal tract. Action: Potent **depressant**; slows bodily functions; powerful analgesic. Consumed by snorting or intravenous injection.
- **2. Cannabinoids:** Derived from inflorescences, leaves, flower tops, and resin of Indian hemp (*Cannabis sativa*). Produce marijuana, hashish, charas, and ganja. Mode of action: Interacts with **cannabinoid receptors in the brain**. Action: Distorts perception and motor coordination; markedly **alters the cardiovascular system** (tachycardia). Inhaled or ingested orally.
- **3. Coca Alkaloids (Cocaine / Coke / Crack):** Extracted from leaves of South American coca plant (*Erythroxylum coca*). Mode of action: Interferes with the reuptake of the neurotransmitter **Dopamine**. Action: Potent CNS **stimulant** inducing euphoria, alertness, and increased stamina; excessive doses cause severe paranoia and **hallucinations**.